

Team Three Standard Deviations

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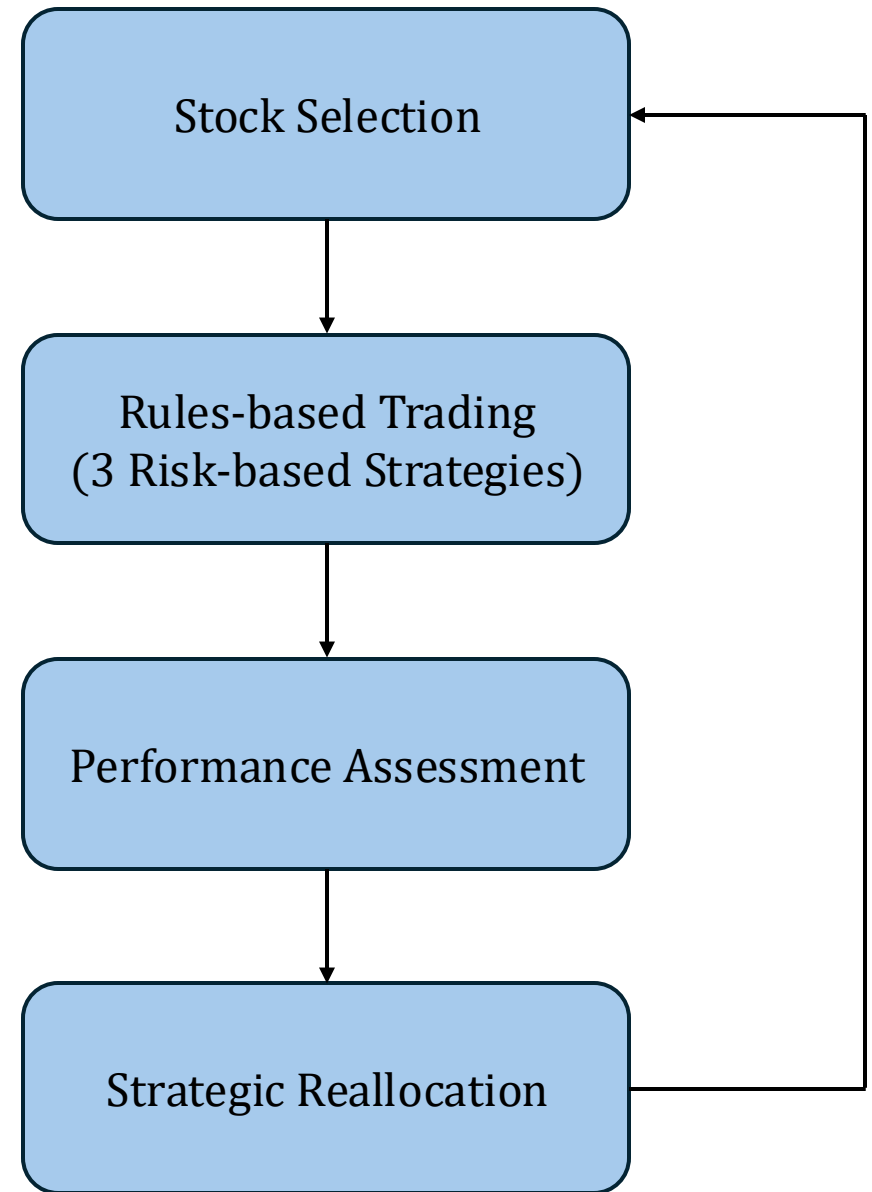
Version 1.0



Algorithmic Trading Flow Chart

Goal

Build a comprehensive, rules-based trading system with dynamic stock selection and strategic reallocation, designed to adapt to changing market regimes and stock movements to maximize returns.



1) Stock Selection

Step 1: Exclude **illiquid** or **overly volatile** stocks to ensure stable, tradable candidates. Reject stocks if:

1. 28 – day annualized volatility > 50%, or
2. 28 – day average daily volume < 100,000 shares

Step 2:

$\hat{Y}_{t+k}^{(i)} = f_{XGB}(X_t^{(i)})$, where the loss minimized is

$$\mathcal{L}^{(i)} = \sum_{t=1}^T \left(Y_{t+k}^{(i)} - \hat{Y}_{t+k}^{(i)} \right)^2 + \lambda \|w\|^2$$

$X_t^{(i)} = [f_1' f_2' \dots f_n']$ is the feature vector at time t for strategy i

$Y_{t+k}^{(i)}$ is the **forward expected return** over the horizon

$\hat{Y}_{t+k}^{(i)}$ is the model's predicted expected return

Stock Selection Rule - Select top N stocks for each strategy with highest predicted expected return

Why?

XGBoost identifies the optimal stocks for each trading strategy by learning complex, non-linear relationships between the stock and the market regime.

Its gradient-boosted decision-tree framework captures subtle patterns that traditional models often miss, enabling more accurate predictions, better feature importance insights, and higher generalization across market regimes.

2) Rules-based Trading (Overview)

Low Risk – Low Reward Strategy

- A signal is generated only when a profit opportunity is almost certain
- Works well in steady, predictable markets

Med Risk - Med Reward Strategy

- A signal is generated only when the stock has shown a predictable movement
- Works well in markets with moderate volatility

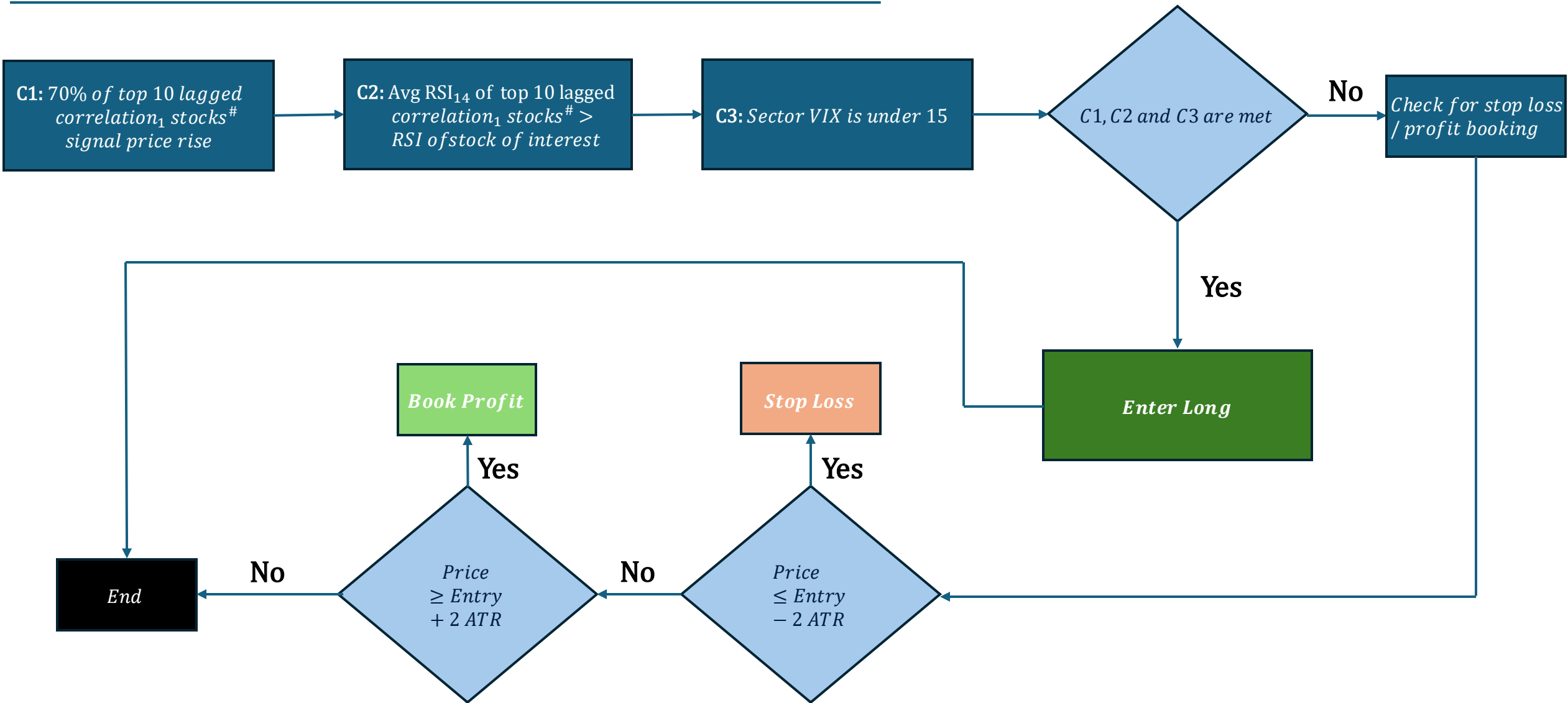
High Risk – High Reward Strategy

- A signal is generated in advance, waiting to seize most opportunities
- Thrives in wildly fluctuating markets, but with moderately predictable movement

Why?

1. Market sentiment can shift rapidly. Relying on a single strategy may lead to poor returns across different market conditions.
2. This three-strategy system maintains a diversified set of strategies, each designed to perform well under specific conditions such as bullish trends, high volatility, range-bound markets, etc.
3. Portfolio weights will be dynamically reallocated based on real-time performance metrics. In **uncertain environments**, the allocation tends to tilt toward **lower-risk** and more **capital-preserving strategies**. When market signals become **clearer**, **higher-return strategies** receive greater weight.

2.1) Strategy 1 **Flowchart** (Low Risk – Low Reward)



Lagged correlation₁ = $\text{Correlation}(p_{it}, p_{jt-1})$,

where p_{it} is the price of the stock of interest at period t , and p_{jt-1} is the price of other stocks j in Nifty 500 at period $t - 1$

2.1) Strategy 1 **Logic** (Low Risk – Low Reward)

Strategy 1 is a low risk – low reward strategy.

How?

This strategy leverages 1 – *day lagged correlations* between stock prices to generate signals, relying on historically strong and predictable co-movements rather than speculative trends.

The 14 – *day RSI* ensures trades are not executed in overbought conditions, while averaging the RSI across the 10 stocks further smooths out noise, protecting against individual stock volatility.

Using *Sector VIX* as a filter avoids sectors with unusually high volatility, limiting exposure to erratic price swings.

The combination of historically reliable signals, multiple confirmations, and volatility screening ensures consistent, low-risk participation in short-term trends, making this strategy suitable for conservative portfolio allocation.

- Stop loss levels are continuously adjusted using a comprehensive **ATR** (*Average True Range*) metric, allowing trades to adapt to changing market volatility. This ensures that positions capture modest upward momentum while locking in profits and limiting losses when price reversals occur.

Note: We have a corresponding set of conditions for taking a short position

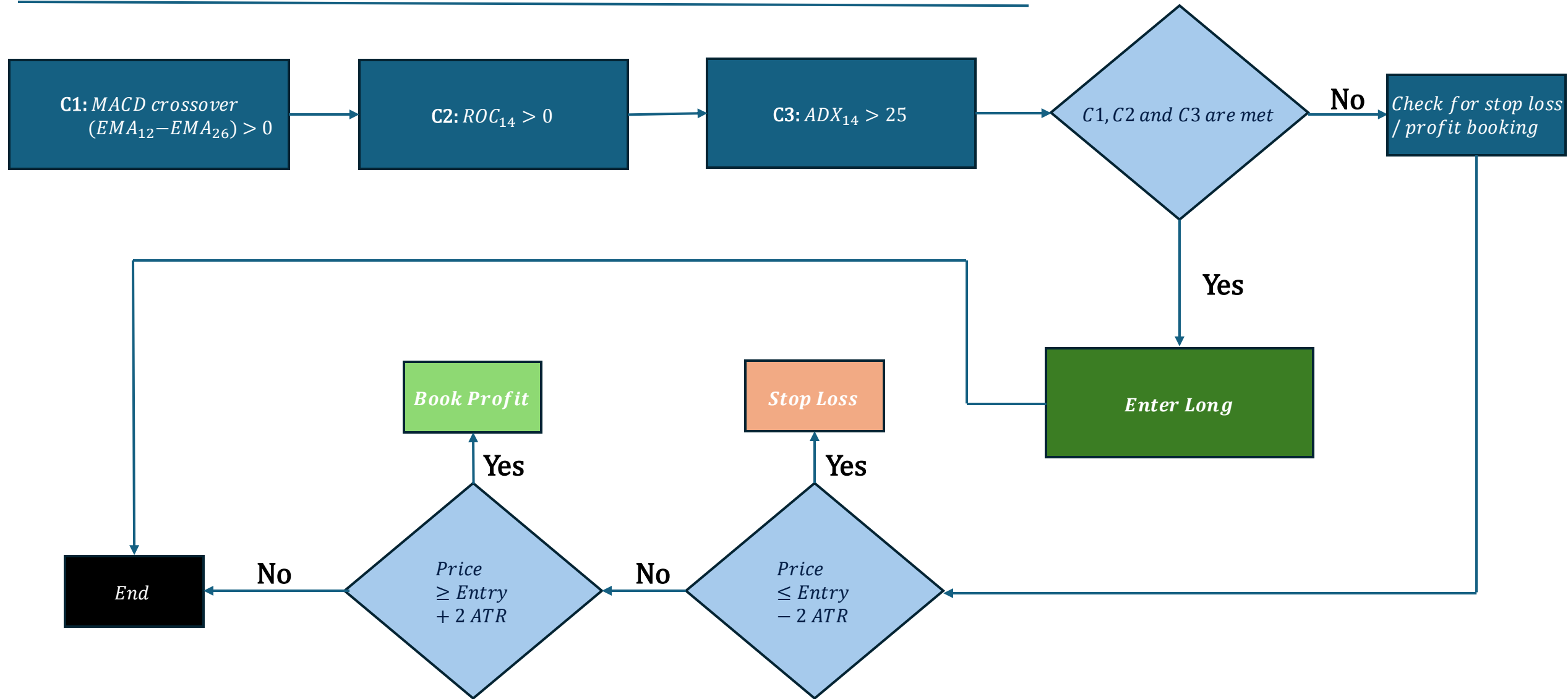
$C1_{short}$ is 70% of top 10 lagged correlated stocks signaling downwards,

$C2_{short}$ is the negation of $C2_{long}$,

$C3_{short}$ remains as is.

Stop loss and profit booking conditions are switched when we take a short position

2.2) Strategy 2 **Flow Chart** (Medium Risk – Medium Reward)



2.2) Strategy 2 **Logic** (Medium Risk – Medium Reward)

Strategy 2 is a medium risk – medium reward strategy

How?

MACD confirms the presence and direction of a trend, reducing exposure to counter-trend trades.

ROC₁₄ (*Rate of Change*) validates the reliability and consistency of that trend, filtering out short-lived or erratic price movements.

ADX₁₄ (*Average Directional Index*) measures trend strength and momentum, ensuring that trades are taken only when sufficient directional force exists to make the position worthwhile.

By layering these confirmations, the strategy balances opportunity and caution, avoiding aggressive entries while still capturing meaningful directional moves, aligning with a medium-risk trading profile.

- We use a dynamic stop-loss mechanism, where stop levels are continuously adjusted based on a comprehensive **ATR** (*Average True Range*) metric. This allows the trade to adapt to changing volatility, capturing extended momentum when the market moves favourably, while also locking in profits or limiting losses when volatility reverses.
- This adaptive risk management approach ensures that the strategy remains responsive yet disciplined, reinforcing its medium-risk profile.

Note: We have a corresponding set of conditions for taking a short position

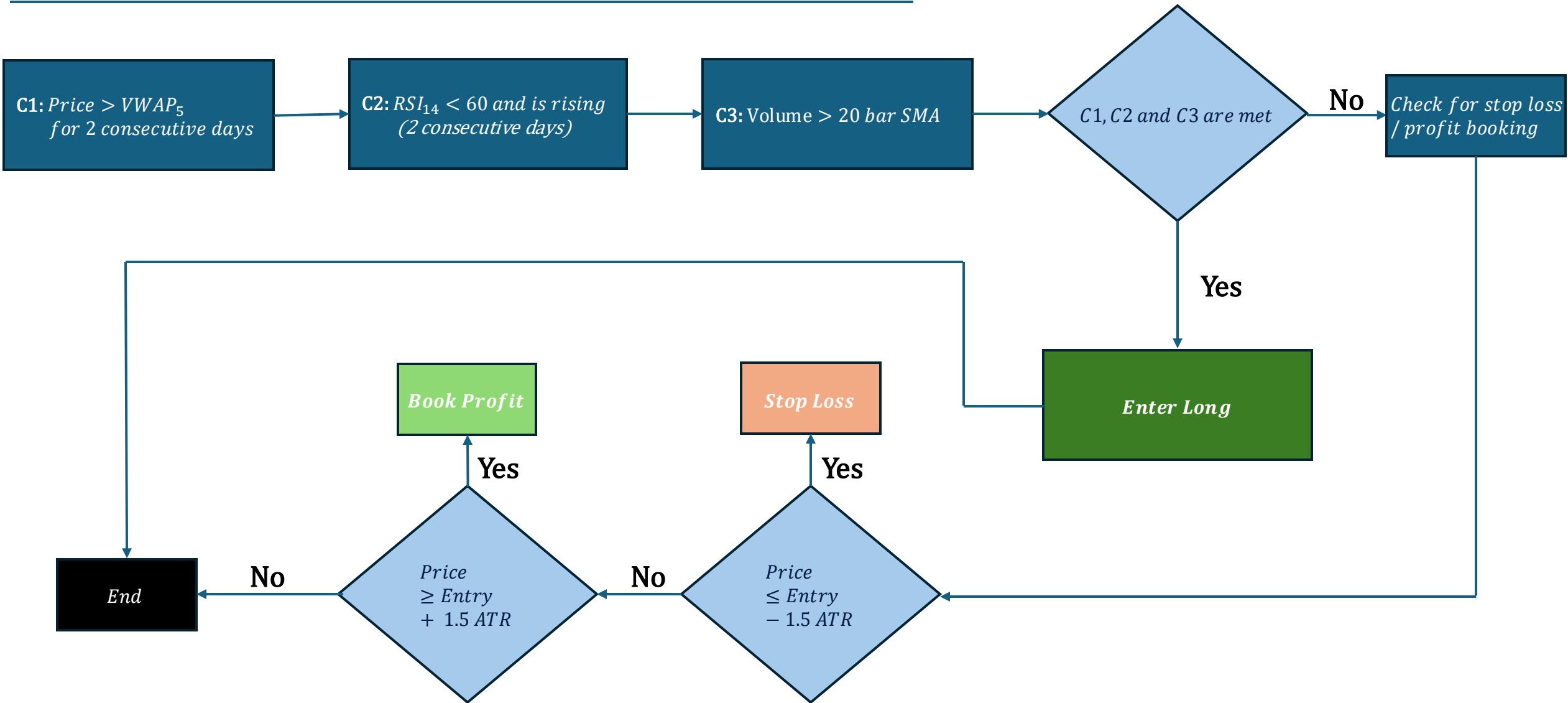
$C1_{short}$ is the opposite of $C1_{long}$,

$C2_{short}$ is $ROC_{14} < 0$,

$C3_{short}$ remains as is.

Stop loss and profit booking conditions are switched when we take a short position

2.3) Strategy 3 **Flow Chart** (High Risk – High Reward)



2.3) Strategy 3 **Logic** (High Risk – High Reward)

Strategy 3 is a high risk – high reward strategy

How?

Price > **VWAP**₅ acts as an early entry trigger, capturing potential upswings before broader confirmation. This early positioning increases profit potential but also heightens exposure to false breakouts.

RSI₁₄ < **60** allows entries even when momentum is moderately strong, ensuring participation in fast-moving trends but leaving limited room for correction if the move reverses.

Volume > **20** – **bar SMA** confirms active participation and short-term enthusiasm conditions that can accelerate gains but also amplify volatility.

In essence, the strategy prioritizes speed and opportunity over safety, seeking to capture sharp, short-lived market moves and hence aligning with a high-risk, momentum-driven trading profile.

- A dynamic stop-loss mechanism continuously adjusts based on the **ATR** (*Average True Range*), allowing the trade to stay aligned with prevailing momentum while automatically tightening when volatility increases.
- The stop-loss scalar is set at **1.5**, tighter than in medium- or low-risk strategies. This ensures **quicker profit capture** and **faster loss containment**, which is essential in a high-risk, momentum-driven environment where price reversals can be abrupt.

Note: We have a corresponding set of conditions for taking a short position

$C1_{short}$ is the negation $C1_{long}$,

$C2_{short}$ is $RSI > 40$ and falling for 2 consecutive days,

$C3_{short}$ remains as is.

Stop loss and profit booking conditions are switched when we take a short position

3) Performance Assessment

$$PAS_{it} = 0.25 * Sharpe_{it}^{14d} + 0.25 * Winrate_{it}^{14d} + 0.2 * Last\ Trade_{it} + 0.15 * Cumulative_{it}^{14d} + 0.15 * (1 - |MDD_{it}|)^{14d}$$

PAS_{it} = Performance Assessment Score of strategy i at time t

$Sharpe_{it}^{14d}$ = Sharpe ratio of the trading strategy i in the last 14 days

$Winrate_{it}^{14d}$ = % of days in the last 14 days where the trading strategy i signalled correctly

$Last\ Trade_{it}$ = Latest returns from the trading strategy i

$Cumulative_{it}^{14d}$ = Cumulative returns of the trading strategy i in the last 14 days

$(1 - |MDD_{it}|)^{14d}$ = $(1 - |Max\ Drawdown|)$ from strategy i in the last 14 days

Why?

- Given the presence of three distinct strategies, it is essential to assess their performance across different market regimes. This evaluation enables optimal portfolio sizing, allowing capital to be allocated based on how each strategy performs under varying conditions.
- The PAS Index serves as a comprehensive performance benchmark, providing a robust basis for portfolio weighting.
- All metrics are expressed on a 0–100 scale, since they are percentage-based, and therefore no additional normalization is required.

4) Portfolio Reallocation

$$Portfolio\ Size_{it} = \frac{e^{PS_{it}}}{\sum_{i=1}^3 e^{PS_{it}}} * \left(\sum_{i=1}^3 Portfolio\ Size_{it-1} \right)$$

Why?

A Softmax-based portfolio reallocation mechanism dynamically adjusts strategy weights based on their PAS scores, assigning higher weights to better-performing strategies in each evaluation period.

This ensures that the portfolio adapts seamlessly to changing market regimes, prioritizing winning strategies while maintaining a non-zero floor allocation for all strategies — allowing underperforming ones to regain exposure when market conditions shift.

Note: If the Softmax-based reallocation proves suboptimal during back-testing, we will explore a Reinforcement Learning–driven reallocation framework to further optimize adaptive weighting.



End of the Presentation